

Exp :18. To classify Iris flowers using the Perceptron algorithm.

Aim

To classify data using the **Perceptron Algorithm** and predict the class label for new samples.

Dataset 1: Fruit Classification

Program

```
from sklearn.linear_model import Perceptron

# Features: Weight, Diameter, Sweetness
X = [
    [150,7.5,90],
    [160,7.8,88],
    [145,7.3,91],
    [120,5.5,70],
    [125,5.8,72],
    [130,6.0,74],
    [155,7.6,89],
    [118,5.4,69],
    [148,7.4,92],
    [122,5.7,71]
]

# Target
y = ["Apple","Apple","Apple","Orange","Orange",
     "Orange","Apple","Orange","Apple","Orange"]

model = Perceptron()
```

```
model.fit(X,y)
```

```
new_sample = [[152,7.5,90]]
```

```
prediction = model.predict(new_sample)
```

```
print("Predicted Fruit:", prediction[0])
```

```
from sklearn.linear_model import Perceptron

# Features: Weight, Diameter, Sweetness
X = [
    [150,7.5,90],
    [160,7.8,88],
    [145,7.3,91],
    [120,5.5,70],
    [125,5.8,72],
    [130,6.0,74],
    [155,7.6,89],
    [118,5.4,69],
    [148,7.4,92],
    [122,5.7,71]
]

# Target
y = ["Apple", "Apple", "Apple", "Orange", "Orange",
     "Orange", "Apple", "Orange", "Apple", "Orange"]

model = Perceptron()
model.fit(X,y)

new_sample = [[152,7.5,90]]
prediction = model.predict(new_sample)

print("Predicted Fruit:", prediction[0])
```

```
... Predicted Fruit: Apple
```

Output

Predicted Fruit: Apple

Result

The Perceptron classifier successfully classified the fruit as **Apple**.

Dataset 2: Student Placement Classification

Program

```
from sklearn.linear_model import Perceptron
```

```
# Features: CGPA, Aptitude Score, Communication Score
```

```
X = [
```

```

[9.3,92,90],
[8.9,87,85],
[8.6,84,82],
[7.3,74,72],
[6.9,69,67],
[9.1,91,89],
[7.4,75,73],
[8.7,85,83],
[6.6,64,60],
[8.8,88,86]
]

y = ["Placed", "Placed", "Placed", "Not Placed", "Not Placed",
     "Placed", "Not Placed", "Placed", "Not Placed", "Placed"]

model = Perceptron()
model.fit(X,y)

new_sample = [[8.8,86,84]]
prediction = model.predict(new_sample)

print("Placement Status:", prediction[0])

```

```

2] from sklearn.linear_model import Perceptron
Os

# Features: CGPA, Aptitude Score, Communication Score
X = [
    [9.3,92,90],
    [8.9,87,85],
    [8.6,84,82],
    [7.3,74,72],
    [6.9,69,67],
    [9.1,91,89],
    [7.4,75,73],
    [8.7,85,83],
    [6.6,64,60],
    [8.8,88,86]
]

y = ["Placed", "Placed", "Placed", "Not Placed", "Not Placed",
     "Placed", "Not Placed", "Placed", "Not Placed", "Placed"]

model = Perceptron()
model.fit(X,y)

new_sample = [[8.8,86,84]]
prediction = model.predict(new_sample)

print("Placement Status:", prediction[0])

```

✓ ... Placement Status: Placed

Output

Placement Status: Placed

Result

The Perceptron classifier successfully predicted that the student is **Placed**.

Dataset 3: Disease Diagnosis

Program

```

from sklearn.linear_model import Perceptron

# Features: Temperature, Heart Rate, Oxygen Level
X = [
    [39.2,112,93],
    [38.8,110,94],
    [37.0,78,99],
    [39.1,114,92],

```

```
[36.8,76,98],  
[38.9,111,94],  
[37.2,80,99],  
[39.3,115,92],  
[36.7,74,98],  
[38.7,109,95]  
]
```

```
y = ["Positive","Positive","Negative","Positive","Negative",  
     "Positive","Negative","Positive","Negative","Positive"]
```

```
model = Perceptron()  
model.fit(X,y)
```

```
new_sample = [[38.9,110,94]]  
prediction = model.predict(new_sample)
```

```
print("Disease Status:", prediction[0])
```

```
from sklearn.linear_model import Perceptron

# Features: Temperature, Heart Rate, Oxygen Level
X = [
    [39.2,112,93],
    [38.8,110,94],
    [37.0,78,99],
    [39.1,114,92],
    [36.8,76,98],
    [38.9,111,94],
    [37.2,80,99],
    [39.3,115,92],
    [36.7,74,98],
    [38.7,109,95]
]

y = ["Positive","Positive","Negative","Positive","Negative",
     "Positive","Negative","Positive","Negative","Positive"]

model = Perceptron()
model.fit(X,y)

new_sample = [[38.9,110,94]]
prediction = model.predict(new_sample)

print("Disease Status:", prediction[0])
```

```
... Disease Status: Positive
```

Output

Disease Status: Positive

Result

The Perceptron classifier successfully diagnosed the disease status as **Positive**.

Dataset 4: Employee Promotion Classification

Program

```
from sklearn.linear_model import Perceptron
```

```
# Features: Experience, Performance Score, Training Hours
```

```
X = [
    [10,95,50],
    [9,92,48],
    [8,90,45],
```

```
[4,72,28],  
[3,68,24],  
[11,96,52],  
[5,75,30],  
[8,89,44],  
[2,60,18],  
[7,87,40]  
]
```

```
y = ["Promoted","Promoted","Promoted","Not Promoted",  
     "Not Promoted","Promoted","Not Promoted",  
     "Promoted","Not Promoted","Promoted"]
```

```
model = Perceptron()  
model.fit(X,y)
```

```
new_sample = [[8,90,46]]  
prediction = model.predict(new_sample)
```

```
print("Promotion Status:", prediction[0])
```

```
from sklearn.linear_model import Perceptron

# Features: Experience, Performance Score, Training Hours
X = [
    [10,95,50],
    [9,92,48],
    [8,90,45],
    [4,72,28],
    [3,68,24],
    [11,96,52],
    [5,75,30],
    [8,89,44],
    [2,60,18],
    [7,87,40]
]

y = ["Promoted", "Promoted", "Promoted", "Not Promoted",
     "Not Promoted", "Promoted", "Not Promoted",
     "Promoted", "Not Promoted", "Promoted"]

model = Perceptron()
model.fit(X,y)

new_sample = [[8,90,46]]
prediction = model.predict(new_sample)

print("Promotion Status:", prediction[0])
```

Output

Promotion Status: Promoted

Result

The Perceptron classifier successfully predicted the employee as **Promoted**.

Dataset 5: Vehicle Classification

Program

```
from sklearn.linear_model import Perceptron
```

```
# Features: Engine Capacity, Weight, Mileage
```

```
X = [
    [100,95,70],
    [125,110,65],
    [150,125,55],
```

```
[800,850,22],  
[1000,950,20],  
[1200,1050,18],  
[110,100,68],  
[900,900,21],  
[140,120,58],  
[1300,1100,17]  
]
```

```
y = ["Bike", "Bike", "Bike", "Car", "Car",  
     "Car", "Bike", "Car", "Bike", "Car"]
```

```
model = Perceptron()  
model.fit(X,y)
```

```
new_sample = [[135,118,60]]  
prediction = model.predict(new_sample)  
print("Vehicle Type:", prediction[0])
```

```
from sklearn.linear_model import Perceptron

# Features: Engine Capacity, Weight, Mileage
X = [
    [100,95,70],
    [125,110,65],
    [150,125,55],
    [800,850,22],
    [1000,950,20],
    [1200,1050,18],
    [110,100,68],
    [900,900,21],
    [140,120,58],
    [1300,1100,17]
]

y = ["Bike", "Bike", "Bike", "Car", "Car",
     "Car", "Bike", "Car", "Bike", "Car"]

model = Perceptron()
model.fit(X,y)

new_sample = [[135,118,60]]
prediction = model.predict(new_sample)

print("Vehicle Type:", prediction[0])
```

Vehicle Type: Bike

Output

Vehicle Type: Bike

Result

The Perceptron classifier successfully classified the vehicle as **Bike**.

Conclusion

The **Perceptron Algorithm** was successfully implemented for classification tasks. It was used to classify: